Project Overview

This project implements an end-to-end machine learning pipeline using the **UCI Heart Disease Dataset** to predict the presence or absence of heart disease. It demonstrates data cleaning, missing value handling, categorical encoding, feature scaling, model training, and evaluation using Python and Scikit-learn.

Objectives

- Download and explore a dataset from the UCI Machine Learning Repository.
- Clean the dataset and handle missing or inconsistent values.
- Apply categorical encoding and numerical feature scaling.
- Train a Logistic Regression classification model.
- Evaluate the model using appropriate classification metrics.
- Visualize the model's performance using graphs.

Dataset

- **Name:** Heart Disease Dataset
- **Source:** UCI Machine Learning Repository
- **Original dataset:** Cleveland subset
- **Records:** Approximately 303
- **Features:** 13 input features
- **Target:** Binary classification (0 = No heart disease, 1 = Heart disease present)

Technologies Used

- Python
- Google Colab
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

Project Workflow

1. Import the required libraries.
2. Download and load the UCI Heart Disease dataset.
3. Explore the data and identify missing values.
4. Clean the data and remove duplicate records.
5. Convert the target into a binary classification variable.
6. Split the data into training and testing sets.
7. Handle missing values using median and most-frequent imputation.
8. Apply One-Hot Encoding to categorical features.
9. Apply StandardScaler to numerical features.
10. Train a Logistic Regression model.
11. Evaluate the model and visualize the results.

Data Preprocessing

- **Missing values:** Numerical missing values are filled using the median, while categorical missing values are filled using the most frequent value.
- **Duplicate records:** Duplicate rows are removed.
- **Categorical encoding:** One-Hot Encoding converts categorical features into numerical representations.
- **Feature scaling:** StandardScaler standardizes numerical features.
- **Train-test split:** 80% training data and 20% testing data.

Preprocessing is implemented within a Scikit-learn pipeline to help prevent data leakage.

Machine Learning Model

Logistic Regression is used to classify whether heart disease is present or absent based on the input features.

Evaluation Metrics

The model is evaluated using:

- **Accuracy:** Proportion of correct predictions.
- **Precision:** Proportion of positive predictions that are correct.
- **Recall:** Proportion of actual positive cases correctly identified.

- **F1-score:** Harmonic mean of precision and recall.
- **ROC-AUC:** Measures the model's ability to distinguish between the two classes.
- **Confusion Matrix:** Summarizes correct and incorrect classifications.

Results and Visualizations

The notebook generates the following results:

- Target class distribution graph.
- Confusion matrix.
- ROC curve.
- Classification report.
- Accuracy, precision, recall, F1-score, and ROC-AUC values.

The numerical results should be taken from the actual execution of the notebook.

How to Run the Project

1. Open Google Colab.
2. Upload or open Assignment_1_ML_Pipeline.ipynb.
3. Run all cells sequentially.
4. Review the evaluation metrics and generated graphs.
5. Save or download the completed notebook.

No local Python installation is required when running the project in Google Colab.

Conclusion

This project demonstrates a complete machine learning workflow, from data acquisition and preprocessing to model training and evaluation. Logistic Regression is evaluated using multiple classification metrics and visualizations. The project provides practical experience in building a reproducible machine learning pipeline using Python and Scikit-learn.

Note: This project is intended for educational purposes and is not a clinically validated medical diagnostic tool.

Author

Machine Learning Assignment – 1